

PARTS LIST

Semiconductors:

Board1 - Arduino Uno board

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1 - 470-ohm

R2 - 82-kilo-ohm

VR1 - 10-kilo-ohm potmeter

Miscellaneous:

CON1 - 11-pin connector

PZ1 - Piezo buzzer

LDR1 - Light-dependent resistor

LCD1 - 16x2 LCD

Software

The program is written in Arduino programming language/sketch. A0 through A5 on Arduino Uno are analogue input signals. The program detects input signals on these pins and prints the corresponding voltage levels on the serial monitor in PC and on LCD1. LDR1 is connected to A0 pin and piezo buzzer to pin 13 of Arduino Uno.

Arduino IDE is used to compile and upload the program to Arduino. Select the correct board from Board→Tools menu in Arduino IDE, select COM port and upload the program (vision.ino)

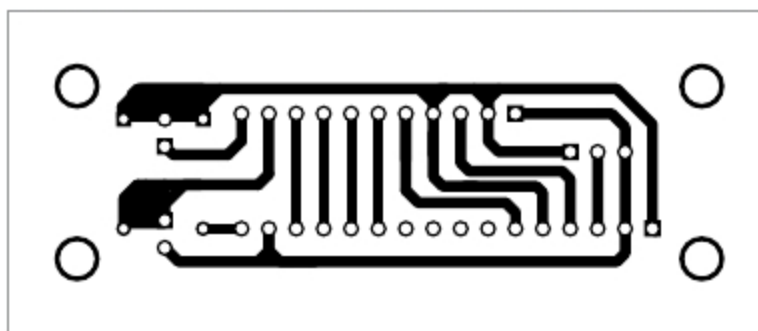


Fig. 2: Actual-size PCB layout of the light-level monitoring system

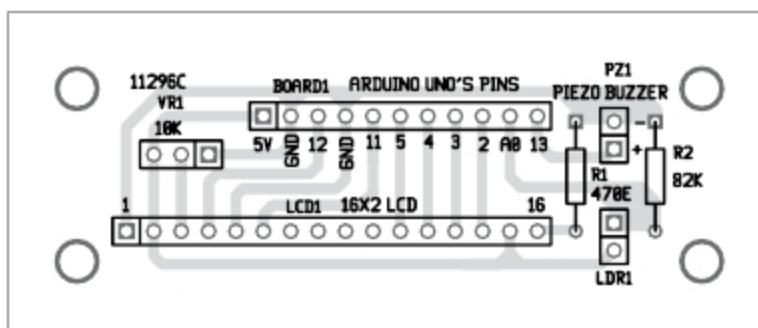


Fig. 3: Components layout for the PCB

through the standard USB port in the computer.

Construction and testing

An actual-size PCB layout of the light-level monitoring system is shown in Fig. 2 and its components layout in Fig. 3. After assembling the circuit,

place the circuit at a location such that light falls on LDR1.

Check connections as per the circuit diagram. Check the light intensity level on LCD1. Whenever there is low light intensity, that is, below twenty per cent, piezo buzzer will sound, and LCD1 will update the status to no vision or dim vision, depending on the light around LDR1 sensor.

If light intensity is above twenty per cent, piezo buzzer will not sound, and LCD1 will show the status as clear

vision or crystal clear, depending on the light around LDR1 sensor. **EFY**



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